JAMES PATTERSON

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PROFESSIONAL SUMMARY

Data scientist with 4 years of experience building machine learning models and analysing complex datasets for financial and retail organisations. I've moved from Python scripts to production pipelines, learned the hard way that data cleaning takes 80% of the time, and genuinely enjoy translating messy business problems into statistical solutions. I'm most interested in predictive modelling and spend my spare time experimenting with time series forecasting.

WORK EXPERIENCE

Senior Data Analyst

DataFlow Solutions

https://dataflow.io

Manchester, United Kingdom

March 2023 - Present

Developed and deployed machine learning models for customer churn prediction, improving retention accuracy to 87% and directly influencing retention strategy for 50,000 active customers.

Built automated data pipelines using Python and Apache Airflow to process 2GB of daily transaction data, reducing manual reporting time by 12 hours per week.

Mentored 2 junior analysts on SQL optimisation and exploratory data analysis, establishing best practices for data validation and documentation.

Created interactive dashboards using Tableau connected to PostgreSQL databases, enabling the marketing team to track campaign performance in real-time across 15 different channels.

Data Scientist

Pinnacle Retail Analytics

https://pinnacle-retail.co

Liverpool, United Kingdom

July 2021 - February 2023

Implemented clustering algorithms using scikit-learn to segment 200,000 retail customers into behavioural groups, enabling targeted marketing campaigns that increased average order value by 18%.

Conducted A/B testing on product recommendation systems, analysing results with statistical significance testing and delivering findings to stakeholders across 6 departments.

Built time series forecasting models in Python to predict inventory demand, reducing stockouts by 23% and excess inventory holding costs by £145,000 annually.

Cleaned and prepared large datasets from multiple sources using Pandas, documenting data quality issues and implementing validation checks within ETL processes.

Data Analyst

Merseyside Financial Group

https://merseyfinance.net

Liverpool, United Kingdom

September 2020 - June 2021

Performed exploratory data analysis on customer transaction data to identify fraud patterns, supporting the risk team in preventing £2.3m in fraudulent transactions.

Created SQL queries to extract and transform data from Oracle databases, serving ad hoc reporting requests for compliance and business intelligence teams.

Developed visualisations in Power BI to communicate findings from statistical analysis to non-technical stakeholders, presenting monthly insights to senior management.

Assisted in the development of a credit scoring model using logistic regression, which improved loan approval accuracy and reduced default rates.

EDUCATION

Bachelor of Science in Mathematics

University of Manchester

Graduated: June 2020

Grade: 2:1

TECHNICAL SKILLS

Programming Languages: Python, SQL, R

Data Analysis and Visualization: Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI

Machine Learning: scikit-learn, TensorFlow, XGBoost, classification, regression, clustering

Databases: PostgreSQL, MySQL, Oracle SQL

Tools and Platforms: Apache Airflow, Git, Jupyter Notebook, Excel

Statistical Methods: Hypothesis testing, A/B testing, time series analysis, correlation analysis